

of e-ink display, then download the library for touch e-ink display. Here we are using the waveshare touch e-ink display, so open Linux and run the following command to get the library and driver for the e-ink display:

```
sudo pip3 install gpiozero
wget http://www.airspayce.com/mikem/
bcm2835/bcm2835-1.68.tar.gz
tar zxvf bcm2835-1.68.tar.gz
cd bcm2835-1.68/
sudo ./configure && sudo make && sudo make
```



Fig. 19: Touch keyboard fixed



Fig. 20: The display fixed



Fig. 21: Laptop running Kali Linux

```
check && sudo make install
sudo apt-get install wiringpi
sudo apt-get update
sudo apt-get install python3-pip
sudo apt-get install python3-pil
sudo apt-get install python3-numpy
sudo pip3 install RPi.GPIO
sudo pip3 install spidev
cd ~
git clone https://github.com/waveshare/
Touch_e-Paper_HAT
```

Go to Example and run the code to show the image on e-ink display. In that code change 'pic' to keyboard layout vector image and run the code (Note: Attach the e-ink touch display with Raspberry on its GPIO pins before running the code) and then turn off the Raspberry. The display will remain with the same image even after power is disconnected, until you refresh the display. E-ink display uses a special technology where the ink molecules that are activated electrically remain on the display, like the printed ink on paper.

Next, create the code that will identify the touch points on the touch e-ink display that we touch and convert those keyboard signals to inputs for Kali Linux in Raspberry Pi. To input signals of mouse and keyboard virtually, without actually connecting any mouse or keyboard, we have many Python libraries. In prototype we used pynput Python library module to give a virtual keyboard input to Raspberry Pi. So, open the Linux terminal and run the following command to install pynput:

```
sudo pip3 install pynput
```

Next, prepare the code that will check the touch points and convert them into key input for each key that we touch on the keyboard.

Import the library and pynput modules for virtual input to Raspberry Pi, as shown in Fig. 13, and check the touch points on display where we will be touching the e-ink display that is being used as a keyboard. Next, create a while loop and define the function to use pynput for the

keyboard inputs to Raspberry Pi, as shown in Fig. 14.

Now we need to change the SPI(0,0) of the Raspberry Pi SPI port to a second SPI. Open the lib folder and make a change in the code file of e-ink touch display by changing to SPI(1,0) in every file where SPI(0,0) is written and save that. Next, enable the second SPI port of Raspberry Pi by adding the line marked by arrow in Fig. 15 in config.txt file of Raspberry Pi in boot folder in root of Raspberry Pi.

Connect the e-ink display to Raspberry Pi pins as shown in Table 2. As we are using only touch part, and touch controller is I2C based, we are connecting the pins of touch controller of e-ink display only to Raspberry Pi.

Preparing the laptop

After setting up everything in the library and modules, and connecting the components and soldering, fix all the hardware in the 3D-printed laptop case that we made earlier. Fix the components as shown in Fig. 17 through Fig. 20. Fit the battery and charging system inside the case, leaving Raspberry Pi 5V pin and GND pin outside.

Testing

Congratulations! The world's smallest hacking laptop is ready to show its magic. Power the laptop and run the keyboard code. You are now ready to use it as a hacking machine that is as powerful as a commercially available hacking laptop, in spite of its tiny size.

You can carry the laptop anywhere for penetration testing, securing network, and then testing the network security. You can also use it, of course, like any other laptop for normal working, playing games, watching videos, or browsing the internet. Whatever you do on your regular laptop, this smallest laptop is capable of doing the same. **EFY**

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